1. An 87-year-old woman with severe dementia is brought to the physician because of fever, chills, lethargy, and agitation for 2 days. Her temperature is 38.3°C (101°F). Examination shows tenderness of the lower abdomen and right costovertebral angle. Laboratory studies show:

Leukocyte count	18,000/mm^3
Segmented neutrophils	65%
Bands	20%
Lymphocytes	15%
Urine	
RBC	20/hpf
WBC	50/hpf
Granular casts	positive
Bacteria	few

Which of the following is the most likely diagnosis?

- A) Allergic interstitial nephritis
- B) Glomerulonephritis
- C) Pyelonephritis
- D) Retroperitoneal abscess
- E) Tubular necrosis

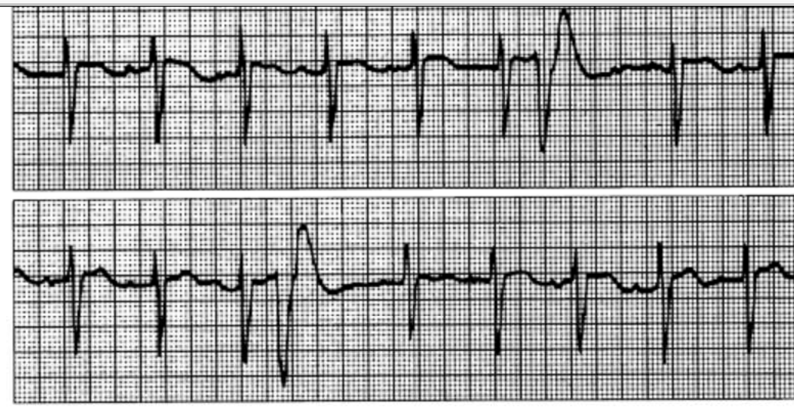
2. A previously healthy 62-year-old man comes to the physician because of a 3-day history of pain with urination and rectal pain and a 1-day history of fever and chills. He has been voiding only small amounts of urine. His temperature is 39.4°C (102.9°F), pulse is 120/min, and blood pressure is 100/70 mm Hg. Examination shows severe tenderness of the prostate. Urinalysis shows nitrites and leukocyte esterase. Which of the following is the most appropriate pharmacotherapy?

- ○ A) Ampicillin
- ○ B) Azithromycin
- ◉ C) Ciprofloxacin
- ○ D) Nitrofurantoin
- ○ E) Tetracycline

3. A 37-year-old woman with chronic renal failure has multiple episodes of hypotension during hemodialysis. Examination shows distended neck veins, clear lung fields, and distant heart tones with no audible murmur or gallop. Echocardiography shows a large pericardial effusion. Which of the following physical signs is associated with the cause of the hypotensive episodes?

- A) Apical systolic murmur
- B) Auscultatory gap
- C) Paradoxical pulse
- D) S_4
- E) Widened pulse pressure

Pericardial effusion = pulsus pardoxus (cardiac tamponade?)



4. Shortly after admission to the hospital for an acute myocardial infarction, a 47-year-old man had the ECG shown. He continues to have chest pain, but his vital signs are stable. The premature beats originate from which of the following?

- A) Atrioventricular reentry pathway
- B) Atrium
- C) Bundle of His
- D) Sinus node
- E) Ventricles

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5. A 52-year-old woman comes to the physician because of left knee pain for 6 months. She has a 10-year history of type 2 diabetes mellitus currently treated with metformin. She has smoked one pack of cigarettes daily for 35 years. She is sexually active and uses no contraception. Her last menstrual period was 2 weeks ago. She has a sedentary lifestyle. She is 163 cm (5 ft 4 in) tall and weighs 138 kg (304 lb); BMI is 52 kg/m^2. Her temperature is 37.3°C (99.2°F), and blood pressure is 150/90 mm Hg. Examination shows no other abnormalities except for an erythematous, edematous left knee. Her nonfasting serum glucose concentration is 200 mg/dL. A joint aspiration is done. Synovial fluid analysis shows:

Appearance	straw colored
Erythrocyte count	0/mm^3
Leukocyte count	1000/mm^3
Segmented neutrophils	90%
Lymphocytes	10%
Crystals	none

An x-ray of the left knee shows narrowing of the medial joint space and subchondral bone sclerosis. Which of the following interventions is most likely to have prevented this patient's condition?

- A) Barrier contraception
- B) Regular exercise
- C) Tight glucose control
- D) Vitamin D and calcium supplementation
- E) Weight reduction

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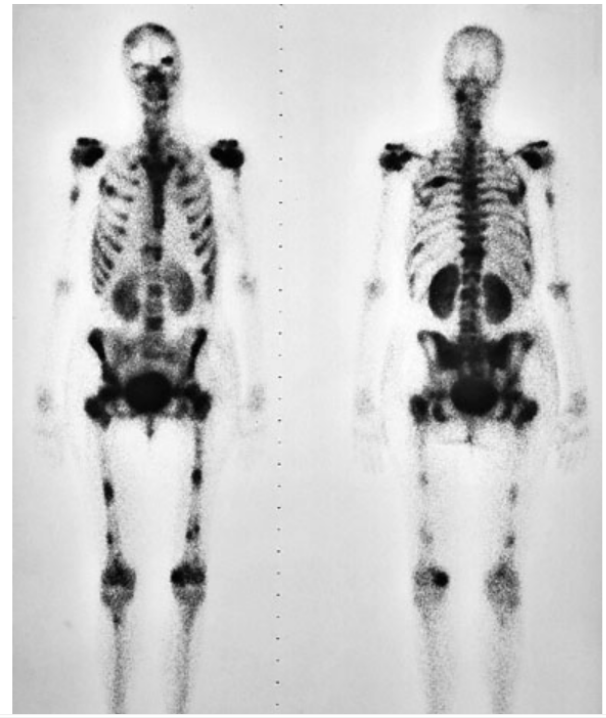
6. A previously healthy 57-year-old man comes to the physician because of a 1-month history of increasing pain at multiple sites in his back, arms, and legs. Examination shows no abnormalities. A bone scan is shown. Which of the following is the most likely diagnosis?

- (•) A) Bone metastasis
- ( ) B) Hyperparathyroidism
- ( ) C) Osteoarthritis
- ( ) D) Osteomyelitis
- ( ) E) Osteoporosis



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7. A 62-year-old white man comes to the physician because of an 8-month history of progressive pain and stiffness of his hands. The stiffness is worse at the end of the day. He has a 1-year history of fatigue and increased urination. He has no history of serious illness and takes no medications. His last visit to a physician was 10 years ago. He does not smoke or drink alcohol. He is 185 cm (6 ft 1 in) tall and weighs 82 kg (180 lb); BMI is 24 kg/m^2. His pulse is 84/min, and blood pressure is 136/82 mm Hg. Examination shows dark brown skin. S_1 and S_2 are normal. An S_3 is heard at the apex. There is mild tenderness over the second and third metacarpophalangeal joints bilaterally without synovial thickening. Heberden nodes are present over the distal interphalangeal joints of the index and ring fingers bilaterally. Laboratory studies show:

Hemoglobin	16 g/dL
Leukocyte count	7700/mm^3
Platelet count	332,000/mm^3
Serum	
Glucose	182 mg/dL
Albumin	3.4 g/dL
Total bilirubin	1.1 mg/dL
Alkaline phosphatase	52 U/L
AST	55 U/L
ALT	68 U/L
Hepatitis B surface antigen	negative
Hepatitis C antibody	negative
Rheumatoid factor	negative

Which of the following is most appropriate to confirm the diagnosis?

Which of the following is most appropriate to confirm the diagnosis?

- A) Human leukocyte antigen typing
- B) Measurement of serum copper concentration
- C) Measurement of serum ferritin concentration
- D) Serum antinuclear antibody assay
- E) Serum rheumatoid factor assay

8. A 77-year-old woman with a 7-year history of hypertension comes for a follow-up examination. She has had intermittent atrial fibrillation despite cardioversion 2 months ago. She has no history of cerebral infarction or transient ischemic attack. Current medications include hydrochlorothiazide and enalapril. Her temperature is 37°C (98.6°F), pulse is 104/min and irregularly irregular, respirations are 14/min, and blood pressure is 170/90 mm Hg. Examination shows no other abnormalities. Which of the following is the most appropriate pharmacotherapy to prevent cerebral infarction in this patient?

- ○ A) Aspirin
- ○ B) Clopidogrel
- ○ C) Dipyridamole
- ● D) Warfarin
- ○ E) No pharmacotherapy indicated

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9. A previously healthy 48-year-old man has the sudden onset of unremitting, severe chest pain on the left that radiates to the left arm and back. He does not smoke cigarettes or drink alcohol. His temperature is 36.8°C (98.2°F), pulse is 115/min and regular, and blood pressure is 90/60 mm Hg. A grade 2/6 diastolic murmur is heard in the upper sternal area. Arterial blood gas analysis shows normal findings. An ECG shows nonspecific ST-T wave changes, and an x-ray of the chest shows a small left pleural effusion. Which of the following is the most likely diagnosis?

- (●) A) Aortic dissection
- ( ) B) Bacterial endocarditis
- ( ) C) Pulmonary embolism
- ( ) D) Ruptured pulmonary bleb
- ( ) E) Unstable angina pectoris

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10. A 38-year-old woman comes to the physician because of a low-grade fever and generalized rash for 4 days. She is currently receiving cefazolin therapy for chronic osteomyelitis. Her temperature is 38.2°C (100.8°F), pulse is 100/min, and blood pressure is 150/108 mm Hg. Examination shows a faint diffuse maculopapular rash. Examination of the back shows no costovertebral angle tenderness. Cardiac and pulmonary examinations show no abnormalities. Laboratory studies show:

Leukocyte count	10,800/mm^3
Segmented neutrophils	60%
Bands	8%
Eosinophils	4%
Lymphocytes	20%
Monocytes	8%
Serum	
Urea nitrogen	20 mg/dL
Creatinine	1.6 mg/dL
Urine	
WBC	12/hpf
RBC	8/hpf
RBC casts	none
WBC casts	rare

Eosinophils are found in the urine sediment. Which of the following is the most likely location of the patient's lesion?

- A) Glomerulus
- B) Renal afferent arteriole
- C) Renal artery
- D) Renal efferent arteriole
- E) Renal tubule

11. A 67-year-old woman comes to the physician for a follow-up examination 2 weeks after undergoing a sigmoid colectomy for perforated diverticulitis. She has had fever for 2 days. Current medications include levofloxacin and metronidazole. Her temperature is 38.5°C (101.3°F), pulse is 80/min, respirations are 16/min, and blood pressure is 120/80 mm Hg. The lungs are clear to auscultation and percussion. Abdominal examination shows a well-healing surgical wound. Laboratory studies show:

Leukocyte count	8400/mm^3
Segmented neutrophils	60%
Eosinophils	15%
Basophils	5%
Lymphocytes	15%
Monocytes	5%
Urine	
Specific gravity	1.015
RBC	2/hpf
WBC	none
Leukocyte esterase	none

A CT scan of the abdomen shows normal postoperative changes. Which of the following is the most likely cause of the fever?

- A) Atelectasis
- B) Drug allergy
- C) Intra-abdominal abscess
- D) Pneumonia
- E) Urinary tract infection
- F) Wound infection

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12. A 67-year-old woman with hypertension comes to the physician because of severe pain and swelling of her right great toe since awakening this morning. The pain has persisted, and she is now unable to bear weight on her right foot. There is no history of trauma. She has never smoked, and she drinks alcohol only on social occasions. Her medications are hydrochlorothiazide and 325-mg aspirin. She is 170 cm (5 ft 7 in) tall and weighs 90 kg (198 lb); BMI is 31 kg/m^2. Her temperature is 36.8°C (98.3°F), pulse is 78/min, respirations are 12/min, and blood pressure is 146/82 mm Hg. Examination shows erythema and swelling of the first metacarpophalangeal joint. An x-ray of the right foot shows soft tissue swelling. Microscopic examination of aspirated joint fluid shows sodium urate crystals. Which of the following is the most appropriate next step in pharmacotherapy?

- A) Acetaminophen
- B) Allopurinol
- C) Indomethacin
- D) Prednisone
- E) Probenecid

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13. A 72-year-old woman is brought to the emergency department because of a 12-hour history of chest tightness, shortness of breath, orthopnea, and paroxysmal nocturnal dyspnea. She has type 2 diabetes mellitus, hypertension, and coronary artery disease. Medications include glyburide, atenolol, and aspirin. She is having difficulty speaking because of moderate respiratory distress. Her pulse is 104/min and regular, respirations are 28/min, and blood pressure is 168/100 mm Hg. Examination shows jugular venous distention with the patient sitting at a 90-degree angle. Crackles are heard at both lung bases. Cardiac examination shows an S_3 gallop. Arterial blood gas analysis on an Fio_2 of 40% shows:

pH	7.48
Pco_2	30 mm Hg
Po_2	102 mm Hg

An x-ray of the chest shows fluffy bilateral interstitial and alveolar infiltrates. Which of the following is the most likely mechanism of the dyspnea?

- (●) A) Alveolar-arteriolar mismatch
- ( ) B) Central hypoventilation
- ( ) C) Decreased alveolar macrophage activity
- ( ) D) Decreased inspired oxygen
- ( ) E) Increased parenchymal elasticity
- ( ) F) Increased tissue oxygen demand
- ( ) G) Large airway obstruction
- ( ) H) Neuromuscular dysfunction
- ( ) I) Oxygen toxicity
- ( ) J) Smooth muscle contraction

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14. A 67-year-old woman comes to the physician because of a 2-day history of passing 12 watery stools daily. She has no other history of abnormal bowel function. Two weeks ago, she was treated in the hospital with antibiotic therapy for right lower lobe pneumonia. She had a myocardial infarction 5 years ago. She underwent a total abdominal hysterectomy 18 years ago for bleeding leiomyomata uteri. She has a 20-year history of rheumatoid arthritis. Current medications include prednisone and aspirin. She appears lethargic. Her temperature is 39.1°C (102.4°F), pulse is 120/min, respirations are 18/min, and blood pressure is 98/68 mm Hg. Examination shows decreased skin turgor. Abdominal examination shows distention and right lower quadrant tenderness with voluntary guarding. Laboratory studies show:

Hemoglobin	11.8 g/dL
Hematocrit	36%
Leukocyte count	18,900/mm^3
Platelet count	325,000/mm^3
Serum	
Na^+	149 mEq/L
K^+	3.1 mEq/L
Cl^-	90 mEq/L
HCO_3^-	32 mEq/L

Flexible sigmoidoscopy shows pseudomembranes. Which of the following is the most likely cause of this patient's current symptoms?

A) Autoimmunity
B) ~~Bowel ischemia~~ ?
C) Electrolyte imbalance
D) ~~Inflammatory bowel disease~~
E) ~~Ingested spores~~
F) Perforation of the colon

Electrolyte imbalance. --> would cause decreased potassium (from diarrhea) and hypotension from fluid loss.

15. A previously healthy 72-year-old woman is brought to the emergency department 1 hour after being found on her living room floor by neighbors. On arrival, she is alert but lethargic. She has pain in her right hip. She says that she tripped and fell 2 days ago and was unable to get up. Her pulse is 140/min and regular, and blood pressure is 88/54 mm Hg. Examination shows dry mucous membranes. The axillary area is dry. The right lower extremity is shorter than the left and is held in external rotation. Laboratory studies show:

Hemoglobin	17.5 g/dL
Leukocyte count	14,500/mm^3
Serum	
Na^+	155 mEq/L
Glucose	295 mg/dL

An x-ray of the right hip shows a fracture of the femoral neck. The most appropriate next step is administration of which of the following?

- A) Intravenous 5% dextrose in water only
- B) Intravenous 5% dextrose in water and subcutaneous insulin
- C) Intravenous dopamine
- D) Intravenous 0.45% saline
- E) Intravenous 0.9% saline
- F) Subcutaneous insulin only

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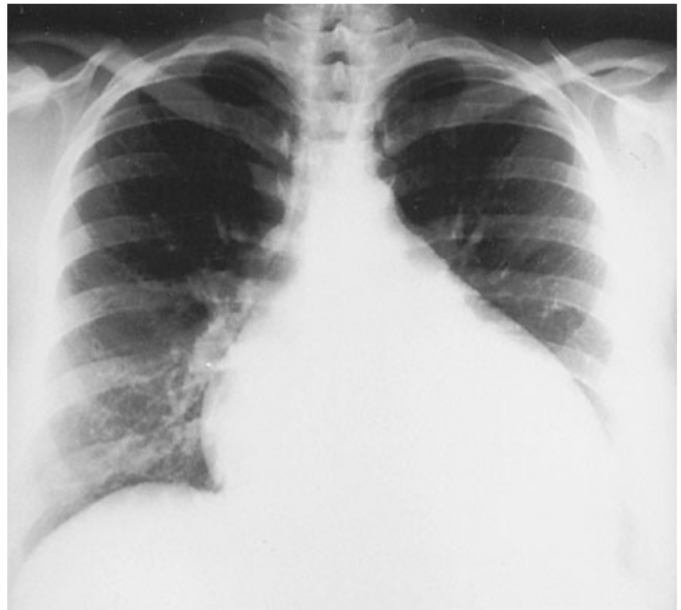

16. A 56-year-old man has had increasing shortness of breath on exertion over the past 6 months. Examination shows mild edema of the lower extremities. Bilateral basilar crackles are heard on auscultation. An x-ray of the chest is shown. Echocardiography shows dilated left and right ventricles. Which of the following is the most likely diagnosis?

- A) Atrial septal defect
- B) Cardiomyopathy (selected)
- C) Chronic pulmonary emboli
- D) Cor pulmonale
- E) Rhabdomyosarcoma

17. A 72-year-old woman comes to the physician because of a 3-month history of shortness of breath, tightness in her chest, and light-headedness that occur during exercise. Her pulse is 80/min and regular, respirations are 18/min, and blood pressure is 120/70 mm Hg while supine. The lungs are clear to auscultation. A grade 4/6 systolic murmur is heard best at the left sternal border. Echocardiography shows left ventricular wall thickening and a calcified aortic valve with an area of 0.8 cm^2. Cardiac catheterization confirms aortic valve disease and shows no evidence of coronary artery disease. Which of the following is the most appropriate next step in management?

- ○ A) Observation only
- ○ B) Digoxin therapy
- ○ C) Propranolol therapy
- ○ D) Warfarin therapy
- ◉ E) Aortic valve replacement

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18. One day after admission to the hospital for treatment of a right femoral fracture, a 77-year-old woman develops dyspnea and confusion. Her pulse is 112/min, respirations are 24/min, and blood pressure is 144/76 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 85%. She is not oriented to place or time. Examination of the skin shows petechiae over the face, neck, and thorax. Scattered rhonchi are heard with no crackles or wheezes. Her platelet count is 310,000/mm^3. Which of the following is the most likely diagnosis?

- A) Aspiration pneumonia
- B) Deep venous thrombosis with pulmonary embolism
- C) Disseminated intravascular coagulation
- D) Fat embolism
- E) Tension pneumothorax

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19. An unconscious 25-year-old man is brought to the emergency department 15 minutes after his neighbor found him lying in the street. The neighbor reports that the patient was jogging before falling to the ground. The patient is wearing a medic alert bracelet that states that he has type 1 diabetes mellitus. No other history can be obtained. There is no evidence of trauma. He does not respond to verbal stimuli but moans and thrashes in response to painful stimuli. His temperature is 37°C (98.6°F), pulse is 90/min, respirations are 12/min, and blood pressure is 122/70 mm Hg. Examination shows no other abnormalities. Which of the following is the most appropriate next step in management?

A) ~~Urine toxicology screening~~
B) ~~MRI of the brain~~
C) Administration of a bolus of insulin
D) ~~Administration of 50% dextrose in water~~
E) Rapid infusion of 1 L of 0.9% saline
F) ~~Lumbar puncture~~

E) NOT DKA. Jogging --> hypoglycemic event. Another answer could be a glucagon pen.

20. A 19-year-old woman comes to the physician because she has never had a menstrual period. On questioning, she states that she has also had a lifelong inability to differentiate smells. Breast development is Tanner stage 2. There is sparse pubic and axillary hair. Pelvic examination shows a normal-appearing cervix. Which of the following is the most likely diagnosis?

- A) Craniopharyngioma
- B) Delayed puberty
- C) Growth hormone deficiency
- D) Hypogonadotropic hypogonadism (Kallmann syndrome)
- E) Pituitary adenoma

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21. An African American husband and wife come to the physician for advice concerning their risk for having children with sickle cell disease. The wife has hemoglobin A on screening; the husband has not been screened. This couple's risk for having offspring with sickle cell disease is closest to which of the following?

- (•) A) 0%
- ( ) B) 25%
- ( ) C) 50%
- ( ) D) 75%
- ( ) E) Unable to be determined without screening of the husband

22. A previously healthy 82-year-old man comes to the physician because of a 2-month history of decreased energy. He has had a 3.2-kg (7-lb) weight loss during this period due to loss of appetite. He appears mildly emaciated. He is 170 cm (5 ft 7 in) tall and weighs 60 kg (132 lb); BMI is 21 kg/m^2. His temperature is 37°C (98.6°F), pulse is 68/min, and blood pressure is 90/50 mm Hg. The lungs are clear to auscultation. Cardiac examination shows no murmurs. Laboratory studies show:

Hemoglobin	14 g/dL
Leukocyte count	7500/mm^3
Segmented neutrophils	56%
Eosinophils	23%
Lymphocytes	14%
Monocytes	7%
Serum	
Na^+	128 mEq/L
K^+	6 mEq/L
Urea nitrogen	30 mg/dL
Creatinine	1.6 mg/dL

Which of the following is the most appropriate next step in diagnosis?

- (●) A) ACTH stimulation test
- ( ) B) Three serial tests of the stool for ova and parasites
- ( ) C) Measurement of urine pH
- ( ) D) Blood culture
- ( ) E) HIV antibody test
- ( ) F) Renal ultrasonography
- ( ) G) Bone marrow biopsy

23. A 37-year-old man comes to the physician because of a 3-month history of intermittent fever and a nonproductive cough. He has had a 14-kg (30-lb) weight loss during this period. Two months ago, an ophthalmologist treated him for left anterior uveitis. Examination shows no abnormalities. His leukocyte count is within the reference range. Serum studies show an AST activity of 100 U/L and alkaline phosphatase activity of 200 U/L. X-rays of the chest show hilar adenopathy. Diffusion capacity for carbon monoxide is 70% of predicted. A PPD skin test is negative. Which of the following is the most likely diagnosis?

- A) Brucellosis
- B) Cytomegalovirus infection
- C) Non-Hodgkin lymphoma
- D) Sarcoidosis
- E) Tuberculosis

Sarcoidosis. Anterior uveitis, sarcoidosis can cause decreased DLCO later in disease.

24. A 62-year-old man is brought to the emergency department 24 hours after the onset of fever, right-sided abdominal pain, and confusion. He has no history of serious illness. Current medications include daily aspirin. He drinks one glass of wine daily. On arrival, he is oriented to person but not to place or time. He becomes light-headed on sitting up. His temperature is 38.7°C (101.7°F), pulse is 120/min, and blood pressure is 84/50 mm Hg while supine. Abdominal examination shows right lower quadrant tenderness with guarding and rebound; there is a suggestion of a mass. The upper and lower extremities are cool and clammy. Urinalysis shows no abnormalities. Additional laboratory studies show:

Leukocyte count	25,400/mm^3 with a shift to the left
Serum	
Na^+	140 mEq/L
K^+	4.5 mEq/L
Cl^-	103 mEq/L
HCO_3^-	19 mEq/L
Urea nitrogen	40 mg/dL
Creatinine	1.6 mg/dL

Arterial blood gas analysis on room air shows:

pH	7.2
Pco_2	34 mm Hg
Po_2	84 mm Hg

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Which of the following serum concentrations is most likely to be increased in this patient?

- A) Acetone
- B) Alcohol
- C) Ethylene glycol
- D) Glucose
- E) Lactic acid
- F) ~~Salicylate~~

25. A 22-year-old man with a history of asthma comes to the emergency department 3 hours after the acute onset of wheezing that did not respond to inhalation of a β-adrenergic agonist. He has had a sore throat, nasal congestion, and a mild cough for 3 days. Two hours after administration of a β-adrenergic agonist by high-flow nebulizer and intravenous fluids, the lungs are clear to auscultation, and his FEV_1 has increased from 40% to 90% of predicted. Which of the following is the most effective pharmacotherapy to prevent a relapse during the next week?

- A) Oral antibiotics
- B) Oral corticosteroids
- C) Oral theophylline
- D) Inhaled cromolyn sodium
- E) Inhaled ipratropium bromide

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Exam Section : Item 26 of 50
Mark

National Board of Medical Examiners
Medicine Self-Assessment

Time Remaining:
3 hr 39 min 47 sec

26. A 77-year-old woman is brought to the emergency department 30 minutes after the sudden onset of an inability to speak and right-sided paralysis while eating breakfast. She has hypertension, type 2 diabetes mellitus, and hypercholesterolemia. She grunts in response to noxious stimuli and is unable to follow commands. Her temperature is 37°C (98.6°F), pulse is 80/min and regular, and blood pressure is 140/90 mm Hg. Examination shows a strong left gaze preference, a right lower facial droop, and right hemiplegia. A CT scan of the head obtained within 2 hours of stroke onset shows effacement of the left hemispheric sulci, consistent with early mass effect due to a left anterior and middle cerebral artery infarction. Conservative management is recommended by the physician due to the size of the stroke. Her son, who is the patient's health care proxy, wants his mother to receive the "clot buster" (recombinant tissue plasminogen activator [rt-PA]) despite the physician's recommendations. When told that rt-PA is medically contraindicated because of the size of the stroke, he says, "I don't care. My mother would never want to be a vegetable, and if she dies from the clot buster, that's better than living paralyzed in a nursing home!" Which of the following is the most appropriate next step in management?

- ○ A) Consult with the hospital ethics committee concerning administration of rt-PA
- ○ B) Consult with the hospital lawyer concerning administration of rt-PA
- ○ C) Administer rt-PA immediately, and ask another physician to assume care for the patient
- ○ D) Administer rt-PA immediately, because a valid health care proxy is in place
- ◉ E) Do not administer rt-PA despite the son's insistence, because it is medically contraindicated in this setting

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Exam Section : Item 27 of 50
Mark

National Board of Medical Examiners
Medicine Self-Assessment

Time Remaining:
3 hr 39 min 41 sec

27. An asymptomatic 72-year-old woman with a 10-year history of hypertension comes to the physician for a follow-up examination. During the past 6 months, her blood pressure readings have ranged between 160/95 mm Hg and 180/115 mm Hg. Her hypertension was previously well controlled with atenolol. She has a 1-year history of osteoporosis treated with calcium, vitamin D, and alendronate. She does not smoke or drink alcohol. She walks for 30 minutes daily. Both of her parents had hypertension. She is 168 cm (5 ft 6 in) tall and weighs 63 kg (140 lb); BMI is 23 kg/m^2. Her temperature is 37.2°C (99°F), pulse is 64/min and regular, respirations are 12/min, and blood pressure is 180/100 mm Hg in the right arm and 175/95 mm Hg in the left arm. The lungs are clear to auscultation. Cardiac examination shows no murmurs. Carotid pulses are normal. A systolic-diastolic bruit is heard over the abdomen. The remainder of the examination, including funduscopic examination, shows no abnormalities. Serum studies show:

Na^+	138 mEq/L
K^+	3.4 mEq/L
Cl^-	104 mEq/L
HCO_3^-	26 mEq/L
Urea nitrogen	15 mg/dL
Creatinine	1.2 mg/dL

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Urinalysis shows no abnormalities. Which of the following is the most likely cause of this patient's worsening hypertension?

- (•) A) Activation of the renin-angiotensin system
- ( ) B) Coarctation of the aorta
- ( ) C) Hypervolemia secondary to an increased glucocorticoid concentration
- ( ) D) Increased cardiac output
- ( ) E) Increased production of catecholamine
- ( ) F) Nephrocalcinosis
- ( ) G) Vitamin D intoxication

28. A 42-year-old woman comes to the physician because of a 6-month history of mild bloating and diarrhea. She passes eight large, foul-smelling stools daily that are difficult to flush. During the past 3 months, she has had an 11-kg (25-lb) weight loss but has not had any changes in her diet or appetite. She has no history of serious illness and takes no medications. She is not currently in distress. She is 165 cm (5 ft 5 in) tall and weighs 45 kg (100 lb); BMI is 17 kg/m^2. Vital signs are within normal limits. Examination shows mild temporal wasting and a beefy red tongue. Abdominal examination shows no abnormalities. There are scattered ecchymoses and trace edema over the lower extremities. The remainder of the examination shows no abnormalities. Laboratory studies show:

Hematocrit	28%
Mean corpuscular volume	65 μm^3
Leukocyte count	4200/mm^3
Prothrombin time	22 sec (INR=1.5)
Serum	
Albumin	2.5 g/dL
Endomysial IgA antibody	positive

Which of the following is the most likely diagnosis?

- A) Crohn disease
- B) Gluten-sensitive enteropathy
- C) Irritable bowel syndrome
- D) Lactase deficiency
- E) Pancreatic exocrine insufficiency

Exam Section : Item 29 of 50
Mark

National Board of Medical Examiners
Medicine Self-Assessment

Time Remaining:
3 hr 39 min 16 sec

29. A 28-year-old woman is admitted to the hospital because of shortness of breath, wheezing, and cough for 2 days. She has had recurrent episodes of asthma, upper respiratory tract infections, otitis media, and frequent diarrhea since childhood. One year ago, she had a severe anaphylactic reaction when she received a blood transfusion for a bleeding peptic ulcer. Examination shows a prolonged expiratory phase of respiration. Scattered wheezes and rhonchi are heard bilaterally. Her hematocrit, leukocyte count, and platelet count are within normal limits. Which of the following is the most likely diagnosis?

- (●) A) IgA deficiency
- ( ) B) Cystic fibrosis
- ( ) C) Severe combined immunodeficiency
- ( ) D) Thymic-parathyroid dysplasia (DiGeorge syndrome)
- ( ) E) Wiskott-Aldrich syndrome
- ( ) F) X-linked agammaglobulinemia

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30. A 77-year-old woman comes to the physician because of a 1-year history of progressive swelling of the ankles and a 3-month history of shortness of breath with exertion. She has not had chest pain, orthopnea, or paroxysmal nocturnal dyspnea. She takes hydrochlorothiazide for hypertension, verapamil for paroxysmal atrial tachycardia, and levothyroxine for hypothyroidism. Her pulse is 78/min and regular, respirations are 18/min, and blood pressure is 145/72 mm Hg. There is no jugular venous distention. The lungs are clear to auscultation. Examination shows large superficial venous varicosities on the lower extremities and moderate ankle and pedal edema bilaterally. There is loss of hair and mild hyperpigmentation over the legs. Oxygen saturation is 96% at rest and 90% with ambulation. An x-ray of the chest and ECG show no abnormalities. Ventilation-perfusion lung scans show two subsegmental perfusion defects but no ventilation abnormalities. Echocardiography shows mitral annular calcifications. Which of the following is the most likely explanation for this patient's dyspnea?

- A) Cardiac emboli secondary to intermittent arrhythmia
- B) Coronary ischemia
- C) Left ventricular diastolic dysfunction
- D) Mitral insufficiency
- E) Recurrent pulmonary emboli

31. A 46-year-old man has been mechanically ventilated since surgical treatment of a bleeding duodenal ulcer 3 weeks ago. He has a temperature of 39.8°C (103.6°F), pulse of 110/min, and blood pressure of 90/50 mm Hg. His hematocrit is 35%, and leukocyte count is 18,300/mm^3. Cardiac output is 8.9 L/min (N=4–5), and central venous pressure is 15 cm H_2O (N=5–8). He is being aggressively resuscitated with intravenous fluids. Which of the following is the most appropriate next step in management?

A) Administration of dexamethasone
B) Administration of isoproterenol drip
C) Blood transfusions
D) Systemic broad-spectrum antibiotic therapy
E) Total parenteral nutrition

STEPS IN MANAGEMNT OF SEPSIS PATIENT: Patient appears to be in septic shock. 1) fluids, if fluids don't raise the BP then give vasopressors (e.g. epinephrine, if patient). 2) antimicrobials

32. A 42-year-old woman comes to the physician because of a 4-month history of weakness in the right hand and a 2-month history of weakness in the left leg. She has had occasional twitching of muscles in all four extremities that she attributes to nervousness. She has migraines treated with sumatriptan. There is no family history of neurologic disease. Examination shows atrophy and weakness of the hands more on the right than on the left, frequent random twitching in the shoulder girdle muscles, and a left footdrop. Deep tendon reflexes are markedly increased in the extremities. Babinski sign is present on the right. Jaw reflex is brisk. Her speech is slurred. Sensory examination shows no abnormalities. Her serum creatine kinase activity is 335 U/L. Nerve conduction studies show no abnormalities. Electromyography shows acute and chronic denervation in several muscles of both upper extremities and the left lower extremity. Which of the following is the most likely diagnosis?

A) Amyotrophic lateral sclerosis ✓
B) Cervical myelopathy
C) Inclusion body myositis
D) Multiple sclerosis
E) Polymyositis

It is now recognised that CK may be mildly to moderately elevated in patients with amyotrophic lateral sclerosis (ALS).

33. A 37-year-old woman comes to the physician for evaluation of a mole on her left leg because her uncle was recently diagnosed with widely metastatic melanoma. She has had the mole for 15 years, and there have been no recent changes. Examination of the anterior aspect of the left lower extremity shows a 0.5-cm, slightly raised, round, brown nevus with symmetric borders. A 4-mm, smooth, moveable right inguinal lymph node is palpated. Which of the following is the most appropriate next step in diagnosis?

- (●) A) Observation only
- ( ) B) Biopsy of the inguinal lymph node
- ( ) C) Excisional biopsy of the nevus with a rim of normal skin
- ( ) D) Incisional biopsy of the nevus
- ( ) E) Needle biopsy of the nevus

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34. A 25-year-old man who is HIV positive comes to the physician because of a 3-week history of cough and wheezing. He stopped taking his antiretroviral therapy 6 months ago because of adverse effects, and he has had an increasing plasma HIV viral load since then. He has required treatment with prednisone for immune thrombocytopenia over the past 2 months with improvement of his platelet count from 25,000/mm^3 to 100,000/mm^3. His temperature is 38.9°C (102°F), pulse is 95/min, respirations are 30/min, and blood pressure is 100/60 mm Hg. Examination shows shotty bilateral cervical adenopathy. There is dullness to percussion over the left upper posterior thorax, and rhonchi are heard over the same area. Laboratory studies show:

Hematocrit	30%
Leukocyte count	10,000/mm^3
Segmented neutrophils	70%
Bands	5%
Eosinophils	10%
Monocytes	15%
Platelet count	100,000/mm^3

An x-ray of the chest shows a nodular density in the left upper lung field posteriorly. Which of the following is most likely to confirm the diagnosis?

- A) Examination of duodenal aspirate
- B) Examination of the stool for ova and parasites
- ~~C) Toxoplasmosis titers~~
- ~~D) Blood cultures~~
- E) Biopsy and culture of the lung mass → ✓ ?? Watter
- F) Lymph node biopsy

Serologic studies for EBV (Epstein-Barr virus), CMV (cytomegalovirus), HIV, Treponema pallidum, Toxoplasma gondii, or Brucella can be helpful in selected cases.

In patients who may have pulmonary infection with Mycobacterium avium complex (MAC), diagnostic testing includes acid-fast bacillus (AFB) staining and culture of sputum specimens. If disseminated MAC (DMAC) infection is suspected, culture specimens should also include blood and urine.

35. An 82-year-old woman comes to the physician because of constant, increasing abdominal pain for 2 weeks. The pain is not related to eating. She has had generalized itching for the past week. She underwent a cholecystectomy for gallstones at the age of 58 years. She is 168 cm (5 ft 6 in) tall and weighs 79 kg (175 lb); BMI is 28 kg/m^2. Her temperature is 36.8°C (98.2°F), pulse is 76/min, and blood pressure is 142/82 mm Hg. Examination shows jaundice and excoriations on both upper extremities. Abdominal examination shows mild epigastric tenderness. Results of a complete blood count are within the reference range. Serum studies show:

Total bilirubin	11.9 mg/dL
Alkaline phosphatase	305 U/L
AST	34 U/L
ALT	42 U/L
Amylase	56 U/L

Abdominal ultrasonography shows dilation of the common bile duct and pancreatic duct. The pancreas is poorly visualized because of overlying bowel gas. Which of the following is the most appropriate next step in diagnosis?

- A) CT scan of the abdomen
- B) Magnetic resonance cholangiopancreatography
- C) Biliary manometry
- D) Mesenteric angiography
- E) Transhepatic cholangiography

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Time Remaining:
3 hr 38 min 27 sec

36. A 52-year-old woman with breast cancer is brought to the emergency department 8 hours after the onset of temperatures to 39.4°C (103°F), shaking chills, and generalized malaise. She has been receiving chemotherapy via an indwelling central venous catheter for 2 months; her last treatment was 3 weeks ago. She was treated in the hospital 1 month ago for pneumonia. She had diarrhea 2 days ago. Current medications include prochlorperazine as needed, lorazepam, and sertraline. She appears acutely ill. Her temperature is 39.2°C (102.5°F), pulse is 120/min, respirations are 24/min, and blood pressure is 90/50 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 99%. Examination shows no erythema surrounding the catheter site. The lungs are clear to auscultation. A grade 2/6 systolic murmur is heard best at the upper left sternal border without radiation. Laboratory studies show:

Leukocyte count	3200/mm^3
Segmented neutrophils	70%
Bands	10%
Lymphocytes	12%
Monocytes	8%
Urine	
RBC	2/hpf
WBC	2/hpf
Bacteria	occasional

An x-ray of the chest shows no abnormalities. In addition to ceftazidime, empiric antibiotic therapy for this patient should include which of the following medications?

- A) Imipenem
- B) Levofloxacin
- C) Metronidazole
- D) Nafcillin
- E) Vancomycin (selected)
- F) No additional antibiotics are indicated

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37. An 82-year-old man is brought to the emergency department after being found unresponsive in his home by his niece. He has mild hypertension treated with hydrochlorothiazide and type 2 diabetes mellitus controlled by diet. His niece reports that the patient has a 2-year history of mild dementia, Alzheimer type, but has been able to live independently. She went to check on her uncle after she was unable to reach him by phone for the past few days. On arrival, he is obtunded. His temperature is 38.2°C (100.8°F), pulse is 118/min, respirations are 22/min, and blood pressure is 100/60 mm Hg. Examination shows dry mucous membranes and poor skin turgor. There is no jugular venous distention. A few scattered rhonchi are heard in both lung fields. Cardiac examination shows a normal S_1 and S_2 with a grade 2/6, systolic ejection murmur at the aortic area with faint radiation to the neck. The abdomen is soft and nontender; bowel sounds are decreased. There are no masses or organomegaly. Peripheral pulses are decreased. Laboratory studies show:

Hematocrit	48%
Leukocyte count	16,000/mm^3
Segmented neutrophils	70%
Bands	20%
Lymphocytes	9%
Monocytes	1%
Platelet count	240,000/mm^3
Serum	
Na^+	138 mEq/L
Cl^-	95 mEq/L
K^+	4.1 mEq/L
HCO_3^-	26 mEq/L
Urea nitrogen	62 mg/dL
Glucose	840 mg/dL
Creatinine	2.2 mg/dL

Test of the stool for occult blood is negative. An ECG shows no abnormalities. Which of the following is the most likely underlying cause of these findings?

- A) Complete insulin deficiency
- B) Diuretic abuse
- C) Excessive glucagon secretion
- D) Osmotic diuresis
- E) Uremia
- F) Valvular stenosis

38. A 37-year-old woman comes to the physician for a follow-up examination after her blood pressure at a health fair was 152/110 mm Hg. She has mild asthma treated with an albuterol metered-dose inhaler as needed. She has a levonorgestrel IUD in place. There is no family history of cardiovascular disease or hypertension. She does not use illicit drugs. She is 173 cm (5 ft 8 in) tall and weighs 66 kg (145 lb); BMI is 22 kg/m^2. Her blood pressure is 155/108 mm Hg in the right upper extremity and 154/106 mm Hg in the left upper extremity. Examination shows no other abnormalities. Serum studies show:

Na^+	143 mEq/L
K^+	3.1 mEq/L
Cl^-	106 mEq/L
HCO_3^-	28 mEq/L
Mg^{2+}	0.8 mEq/L
Urea nitrogen	8 mg/dL
Glucose	72 mg/dL
Creatinine	0.8 mg/dL

Which of the following is the most likely mechanism of this patient's increased blood pressure?

- A) Adverse effect of levonorgestrel
- B) Cortisol excess
- C) Focal areas of renal ischemia
- D) Mineralocorticoid excess
- E) Norepinephrine, epinephrine, and dopamine excess

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Time Remaining:
3 hr 37 min 29 sec

39. A 40-year-old man who is a health-care worker comes for a follow-up examination after his PPD skin test is positive on annual testing. The test showed 17 mm of erythema and 11 mm of induration at maximum measurement points. Annual PPD skin tests over the past 5 years have been negative. An x-ray of the chest today shows no abnormalities. Which of the following is the most appropriate next step in management?

- A) Administration of BCG vaccine
- B) Administration of isoniazid
- C) Administration of rifampin
- D) No treatment now and repeat PPD skin test
- E) No treatment now and repeat an x-ray of the chest annually

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40. A 52-year-old woman comes to the physician because of a 2-day history of a painful rash on her abdomen. She is currently undergoing chemotherapy for non-Hodgkin lymphoma. Her temperature is 38.5°C (101.3°F). Abdominal examination shows a vesicular rash over the left and right lower quadrants. There are numerous similar lesions over the back. Which of the following is the most appropriate next step in management?

- ○ A) Advise the patient to keep the areas clean
- ◉ B) Intravenous acyclovir therapy
- ○ C) Intravenous broad-spectrum antibiotic therapy
- ○ D) Oral analgesic and diphenhydramine therapy
- ○ E) Topical hydrocortisone therapy

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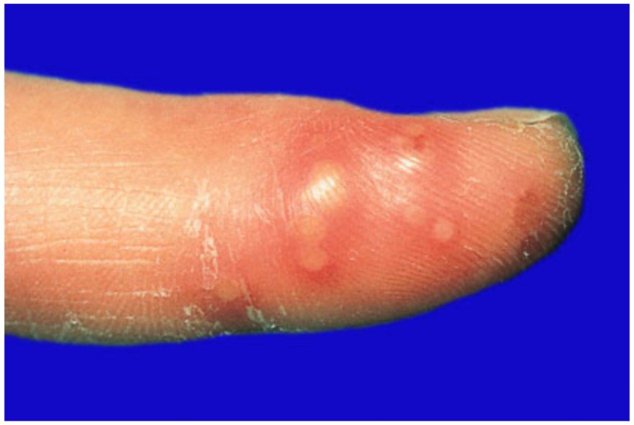
41. A 37-year-old white woman comes to the physician 2 days after noticing painful bumps on her right index finger. She had a similar episode 3 years ago that resolved without treatment. She has no history of serious illness and takes no medications. She is employed as a respiratory therapist. Examination shows tender lesions on the distal phalanx of the right index finger. A photograph of the affected area is shown. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate treatment?



- ○ A) Topical betamethasone therapy
- ◉ B) Oral acyclovir therapy
- ○ C) Oral dicloxacillin therapy
- ○ D) Intravenous cefazolin therapy
- ○ E) Surgical incision and drainage

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Time Remaining:
3 hr 37 min 8 sec

42. An 82-year-old woman is brought to the emergency department by her son after he found her wandering around the house aimlessly. She has lived with her son since the death of her husband 11 years ago. She has a history of mild renal insufficiency. Her serum creatinine concentration is 1.9 mg/dL. The son signs the admitting papers on her behalf and says that he will be traveling to a friend's house 200 miles away tonight. Examination shows no abnormalities other than mild confusion. Laboratory findings are within normal limits. During the night, she suddenly becomes comatose, and a CT scan of the head without contrast is required. Attempts to contact the son are unsuccessful. Which of the following is the most appropriate course of action?

- A) Document in the medical record that the patient had given verbal consent for procedures prior to becoming comatose
- B) Obtain approval from three licensed physicians
- C) Obtain the CT scan without consent
- D) Obtain psychiatric consultation to approve a 72-hour hold for evaluation
- E) Withhold the CT scan pending consent from the son

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43. A 72-year-old man comes to the physician because of a 2-month history of intermittent red-tinged urine. He has not had pain with urination, urinary hesitancy, or flank pain. He has hypertension, emphysema, and lung cancer. Medications include hydrochlorothiazide, potassium, and albuterol. He has smoked one pack of cigarettes daily for 56 years. His temperature is 37°C (98.6°F), pulse is 60/min, and blood pressure is 150/84 mm Hg. Abdominal examination shows no abnormalities. The prostate is firm and diffusely enlarged. Laboratory studies show:

Hematocrit	38%
Mean corpuscular volume	78 μm^3
Platelet count	140,000/mm^3
Serum	
Creatinine	1.4 mg/dL
Prostate-specific antigen	4.3 ng/mL (N<4)
Urine	
Protein	trace
RBC	50–100/hpf
WBC	0–2/hpf
Hyaline casts	2/hpf (N=0–5)
Nitrites	negative

Which of the following is the most likely underlying cause of this patient's hematuria?

Which of the following is the most likely underlying cause of this patient's hematuria?

- A) Bacterial invasion of urogenital epithelium
- B) Deposition of IgA in the renal mesangium
- C) Malignant transformation of epithelial cells
- D) Mechanical irritation of ureteral epithelium
- E) Platelet dysfunction

44. A previously healthy 32-year-old woman comes to the physician because of a 6-week history of shortness of breath, fatigue, and intermittent palpitations. She appears comfortable and well. Her pulse is 84/min, respirations are 14/min, and blood pressure is 120/80 mm Hg. The lungs are clear to auscultation. S_1 is normal, and S_2 is widely split and fixed. A grade 2/6, systolic ejection murmur is heard at the cardiac base. An x-ray of the chest shows right ventricular enlargement, a dilated pulmonary artery, and increased pulmonary vascularity. An ECG shows an incomplete right bundle branch block. Which of the following is the most likely explanation for these findings?

- A) Chronic pressure overload on the right ventricle
- B) Chronic volume overload on the right ventricle
- C) Embolic obstruction of multiple small pulmonary arteries
- D) Fibrotic obstruction of multiple small pulmonary arteries
- E) Pulmonary vasospasm

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45. A 67-year-old man comes to the physician because of four episodes of regurgitating undigested food after meals during the past 4 months. During the past year, he sometimes has had a vague sensation of solid food sticking in his throat and sometimes has heard gurgling when swallowing. He has not had weight loss. He has hypertension and type 2 diabetes mellitus. Current medications include metoprolol, furosemide, and glyburide. He is 170 cm (5 ft 7 in) tall and weighs 104 kg (230 lb); BMI is 36 kg/m^2. The thyroid gland and cervical lymph nodes cannot be palpated due to obesity. Rhonchi are heard over the right lower lobe. Abdominal examination shows central obesity with no tenderness or masses. Which of the following is the most appropriate next step in diagnosis?

- A) Barium swallow
- B) Gastric-emptying scintigraphy
- C) CT scan of the neck
- D) Esophageal manometry
- E) 24-Hour monitoring of esophageal pH

46. A 57-year-old man comes to the physician because of a 3-month history of moderate left leg pain that began at the knee and now involves the entire left leg. He has no history of leg trauma. He has hypertension well controlled with a β-adrenergic blocking agent. His only other medication is acetaminophen as needed, which does not relieve his pain. His blood pressure is 130/80 mm Hg. Examination shows warmth and tenderness to palpation over the left anterior tibia. Sensation and motor function are intact. Distal pulses are 1+. An x-ray of the left leg shows increased cortical thickness along the tibia with slight anterior bowing; there is no periosteal reaction. His serum alkaline phosphatase activity is 100 U/L. A whole-body bone scan shows several areas of increased uptake, including the tibial region. Which of the following is the most likely cause of these findings?

- A) Areas of spindle cells and dysplastic bone
- B) Decreased bone mineralization
- C) Decreased osteoid production
- D) Increased bone turnover
- E) Trabecular microfractures

47. A 52-year-old man with alcoholism is brought to the emergency department because he has been unable to move his lower extremities for the past 4 hours. He is awake and alert. Examination shows slight swelling of the lower extremities. He has oliguria. Bladder catheterization yields only a small quantity of urine that is positive for blood by dipstick. Which of the following is the most likely diagnosis?

- A) Botulism
- B) Cerebral infarction
- C) Conversion disorder
- D) Familial periodic paralysis
- E) Guillain-Barré syndrome
- F) Hypokalemia
- G) Myasthenia gravis
- H) Rhabdomyolysis (selected)
- I) Spinal cord compression

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48. Two days after undergoing surgical repair of a torn anterior cruciate ligament, a 24-year-old man develops jaundice. He has not had fever, nausea, vomiting, or abdominal pain. Examination shows no abnormalities except for mild scleral icterus. Laboratory studies show:

Hemoglobin	16 g/dL
Serum	
Bilirubin, total	3.5 mg/dL
Direct	0.2 mg/dL
Alkaline phosphatase	38 U/L
AST	14 U/L
ALT	12 U/L
Lactate dehydrogenase	120 U/L

For each patient with jaundice, select the most likely diagnosis.

- ○ A) Acute hepatitis
- ○ B) α_1-Antitrypsin deficiency
- ○ C) Biliary atresia
- ○ D) Cholangiocarcinoma
- ○ E) Choledocholithiasis
- ◉ F) Gilbert syndrome
- ○ G) Glucose 6-phosphate dehydrogenase deficiency
- ○ H) Liver abscess
- ○ I) Peptic ulcer disease

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A) ~~Acute hepatitis~~
B) ~~α_1-Antitrypsin deficiency~~
C) Biliary atresia
D) Cholangiocarcinoma
E) Choledocholithiasis
F) ~~Gilbert syndrome~~
G) ~~Glucose 6-phosphate dehydrogenase deficiency~~
H) Liver abscess
I) ~~Peptic ulcer disease~~

49. A 25-year-old woman comes to the physician because of a 5-day history of fatigue, nausea, and decreased appetite. Her temperature is 37°C (98.6°F), pulse is 86/min, and blood pressure is 110/50 mm Hg. She is told that she has a viral infection and is sent home. One week later, she returns because of continued fatigue and jaundice. Her pulse is 80/min, respirations are 12/min, and blood pressure is 110/64 mm Hg. Examination shows scleral icterus. Cardiopulmonary examination shows no abnormalities. The liver edge is palpated 1 cm below the right costal margin and is slightly enlarged, smooth, and tender to palpation. Laboratory studies show:

Hemoglobin	13.2 g/dL
Serum	
Bilirubin, total	4.2 mg/dL
Direct	3.6 mg/dL
Alkaline phosphatase	120 U/L
AST	350 U/L
ALT	280 U/L
Lactate dehydrogenase	410 U/L

LIVER ABSCESS or Cholangiocarcinoma ????????

50. A 32-year-old man comes to the physician for a routine examination. His last office visit was 5 years ago in another community. He underwent an appendectomy 8 years ago. There is no personal or family history of serious illness. Examination shows no abnormalities. Which of the following is the most appropriate screening test for this patient?

- A) ~~Test of the stool for occult blood~~
- B) Measurement of serum cholesterol concentration
- C) Measurement of serum glucose concentration
- D) ~~Complete blood count~~
- E) ECG

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